



P-BLOCK ELEMENTS

NUMERICAL VALUE TYPE QUESTIONS:

1.

Consider the structure of Al_2Me_6 compound and find the value of $\frac{x+z}{3}$

Where x = no of $2c-2e^-$ bonds

z = total no of $3c-2e^-$ bonds

2.

Find the number of properties which is/are correctly related to graphite.

- 1) Most stable allotrope of carbon (thermodynamically)
- 2) Highest electrical resistance among all allotropes of carbon
- 3) It contains dangling bonds
- 4) C – C covalent bond length is higher as compared to diamond.
- 5) football like model
- 6) Hexagonal rings in structure.

3.

The no of elements which are affected by cold H_2O .

C, Si, Ge, Sn, Pb

4.

How many of the following substances / molecules / ion have bond order 1.33?

BF_3 , graphite, NO_3^- , SO_3^{2-} , SO_4^{2-} , PO_4^{3-}

5.

Graphite gives mellitic acid with conc HNO_3 . It supports the hexagonal structure of

graphite. The no of carbon atoms with sp^2 hybridization in mellitic acid is x . Then $\frac{x}{2}$ is.

6.

No of following ions which have distorted octahedral geometry

- i) GeF_6^{2-} ii) $Sn(OH)_6^{2-}$ iii) $[SnCl_6]^{2-}$ iv) $[SiF_6]^{2-}$ v) $[BH_4]^-$

7.

Consider a prototypical fullerene C_{60}
 Let a = Number of 5 – membered rings
 b = Number of 6 – membered rings
 c = Number of π – bonds
 find the value of $(3a - 2b + c)$

8.

Among the given oxides

$B_2O_3, Al_2O_3, In_2O_3, CO_2, SiO_2, CO, SnO_2, PbO_2, SnO, PbO, Tl_2O$

x = Acidic oxides y = Basic oxides, z = Amphoteric oxides m = neutral oxide.

Then $\frac{x + y + z}{m}$ is

9.

Diamond is measured in carats. 4 carats is equal to how many milli grams.

10.

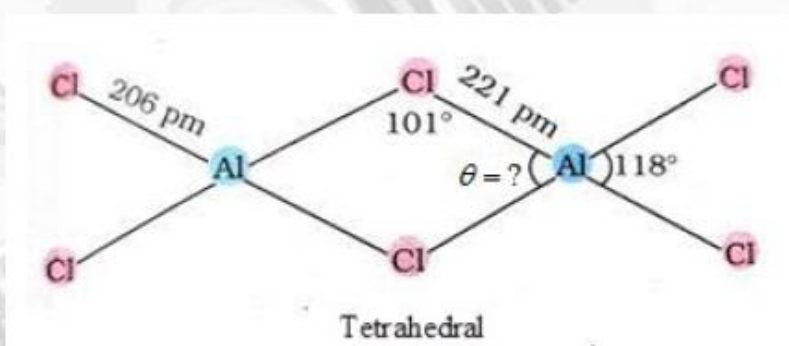
Small C – C bond length in fullere is _____ x pm

C – C bond length in graphite is _____ y pm

$y - x$ is

11.

In the given structure of Al_2Cl_6 find ' θ '



12.

2 moles of Carbon reacts with oxygen un air or oxygen forming two oxides CO and CO_2 depending on the amount of oxygen available. If 19.2 grams of CO is formed. Then % carbon converted into CO is (Atomic weights of $C=12$, $O=16$)

13.

Maximum number of atoms present in a plane in diborane is X.

Maximum number of atoms present in a plane in Borazine is Y.

Then,

what is the value of $\frac{X+Y}{2}$?

(Consider those planes which contain maximum number of atoms)

14.

i). In ortho silicates the number of 'O' atoms shared per tetrahedron is 'a'

ii) In Pyrosilicates the number of 'O' atoms shared per tetrahedron is 'b'

iii) In chain silicates the number of 'O' atoms shared per tetrahedron is 'c'

iv) In three dimensional silicates the number of 'O' atoms shared per tetrahedron is 'd'

v) The Empirical formula of cyclic silicates is $(SiO_y)_n^{2n-}$

What is the value of (a+b+c+d-y) ?

MULTI CORRECT ANSWER TYPE:

15.

Which of the following statements(s) is/are correct?

A) PbI_4 is not known because of the oxidizing power of $Pb(+IV)$ and the reducing power of I^- .

B) $SnCl_2$ does not hydrolyse

C) $SnCl_2$ is reducing agent

D) Tin (II) chloride is linear in shape.

16.

Which of the following is/are correct for group 14 elements?

A) The stability of dihalides are in the order $CX_2 < SiX_2 < GeX_2 < SnX_2 < PbX_2$

B) The ability to form $p\pi - p\pi$ multiple bonds among themselves increases down the group

C) The tendency for catenation decreases down the group

D) They all form oxides with the formula MO_2 .

17.

Correct statements about $AlCl_3$

- A) Aqueous solution of $AlCl_3$ turns red litmus blue
- B) $AlCl_3$ sublimes at $180^\circ C$
- C) In the solid state it has layered structure
- D) Almost above $850^\circ C$, it totally exists as a monomer.

18.

Correct statements among the following

- A) Lapis Lazuli is a precious stone blue in colour
- B) $CuSO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ belongs to pseudo alam
- C) Alums are prepared by using $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 4Al(OH)_3$
- D) Aqueous solutions of alums are acidic in nature.

19.

Wrong statements among the following

- A) Stability of graphite > Diamond > fullerene
- B) graphite, diamond contain dangling bonds
- C) fullerenes can be reduced by Alkali metals formed compounds behave as superconductors
- D) enthalpy of conversion of diamond into graphite is endothermic

20.

$C(OH)_4$ is unstable because a carbon atom cannot hold more than one – OH groups but $Si(OH)_4$ is a stable compound because

- A) C – O bond energy is low
- B) C – O bond energy is high
- C) Si – O bond energy is low
- D) Si – O bond energy is high

21.

Among the following the correct statements is (are)

- A) B_2H_6 contains two bridged bonds between two boron atoms and two hydrogen atoms
- B) $(BeH_2)_n$ has two electron three centered bonds
- C) $PbBr_4$ and PbI_4 cannot exist at room temperature
- D) C_3O_2 has linear structure

22.

What are the products obtained when P_4O_{10} reacts with HNO_3 and $HClO_4$ separately?

- A) N_2O_5
- B) Cl_2O_6
- C) NO_2
- D) Cl_2O_7

MATCHING:

23.

	List-I		List-II
a)	Quartz	p)	Three corners of SiO_4^{-4} tetrahedron shared
b)	Amorphous SiO_2	q)	Piezo electric material
c)	Sheet silicates	r)	Two corners of SiO_4^{-4} are shared
d)	Cyclic silicates	s)	All four corners of SiO_4^{-4} are symmetrically shared
		t)	Corners of SiO_4^{-4} are randomly linked

A) a-r, b-qs, c-p, d-t

B) a-p, b-s, c-r, d-q

C) a-t, b-r, c-p, d-s

D) a-q, s b-t, c-p, d-r

24.

Consider the following list

List – I

A) $\text{S}_2\text{O}_3^{2-}$

B) SO_4^{2-}

C) SO_5^{2-}

D) $\text{S}_2\text{O}_4^{2-}$

List – II

P) S is in sp^3 hybridization

Q) Contain S – S bond

R) Oxidizing agent

S) S reducing agent

A) A – Q; B – PQ; C – PQS; D – R

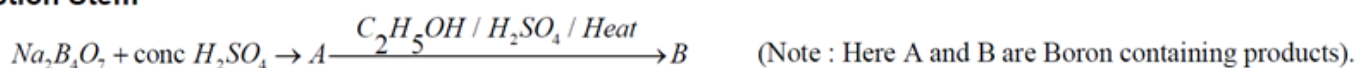
B) A – PQS; B – P; C – PR; D – PQS

C) A – PQ; B – RS; C – PR; D – R

D) A – PQ; B – PR; C – Q; D – RS

PARAGRAPH QUESTIONS:

Question Stem



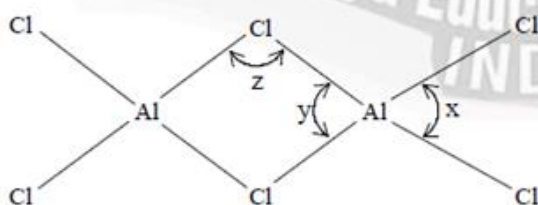
25. The no of B-O-C bonds in compound (B) is

26. (A) on treatment with $\text{H}_2\text{O}_2/\text{OH}^-$ forms a compound 'C' which is used as brightner in washing powder. Find the no.of peroxy bonds present in it.

Paragraph-2:

Anhydrous aluminium chloride exist as a dimer at below 625 K and as a monomer at above 1025 K. In between temperatures it is an equilibrium mixture of AlCl_3 and Al_2Cl_6 .

The structure of dimer is



27. The correct order of bond angles in Al_2Cl_6 is
 A) $x > z > y$ B) $z > x > y$ C) $y > x > z$ D) $z > y > x$
28. Pick out correct statement(s) from the following?
 I. On methylation Al_2Cl_6 forms $(CH_3)_6Al_2$ and B_2H_6 form $(CH_3)_4B_2H_2$
 II. On methylation both Al_2Cl_6 and B_2H_6 forms hexamethyl derivatives $((CH_3)_6M_2)$
 III. In Al_2Cl_6 , Al-Cl-Al bridge is $3c-4e$ bond while in B_2H_6 , B-H-B bridge is $3c-2e$ bond.
 IV. Both Al_2Cl_6 and B_2H_6 acts as Lewis acids
 V. In gaseous state both compounds exist as a mixture of monomer and dimer.
 A) II, III, IV B) I, III, IV C) I, III, IV, V D) II, III, IV, V

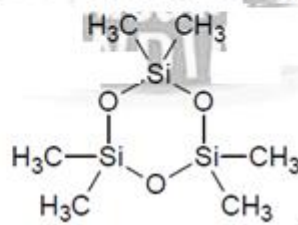
Question Stem



A linear silicone on dehydration forms cyclic silicone. A linear silicone on adding $(CH_3)_3SiCl$ gets terminated.

A linear silicone blocked at both the ends contain four $Si-O-Si$ linkages needed X number of $(CH_3)_2SiCl_2$ units.

For the preparation of cyclic silicone of structure



the number of CH_3MgCl needed is 'Y' according to the above given equation

29. The value of 'X' is _____;
 30. The value of Y is _____

KEY:

1)	8	2)	3	3)	0	4)	4	5)	6
6)	0	7)	26	8)	10	9)	800	10)	2.8-3.8
11)	79	12)	34-35	13)	9	14)	4	15)	AC
16)	ACD	17)	BCD	18)	ABCD	19)	D	20)	AD
21)	ABCD	22)	AD	23)	D	24)	B	25)	3
26)	2	27)	A	28)	B	29)	3	30)	6